

ASSESSX

Product Requirements Document (PRD)

Version: 1.0

Status: Initial Product Definition

Product: AssessX

Category: Secure Assessment, AI Proctoring & Interview Platform

Primary Platform: Windows Desktop + Web Admin Portal

1. PRODUCT OVERVIEW

AssessX is a secure assessment and candidate evaluation platform designed for:

- Universities
- Colleges
- Recruitment companies
- Corporate hiring teams
- Professional certification organizations
- Training institutions

AssessX combines:

1. Secure online examinations
2. AI-powered multimodal proctoring
3. Identity verification
4. Suspicious activity detection
5. Evidence collection
6. Explainable risk scoring
7. Coding assessments
8. Live human interviews
9. AI-powered adaptive interviews
10. Candidate evaluation and reporting

The product should eventually support large-scale concurrent assessments while maintaining low-latency security monitoring.

2. PROBLEM STATEMENT

Traditional remote examinations and interviews face three major problems.

2.1 Assessment Fraud

Candidates may attempt:

- Impersonation
- Smartphone usage
- Additional screens
- Unauthorized people
- Searching the internet
- AI assistance
- Messaging
- Screen sharing
- Remote assistance
- Copy/paste
- Other prohibited behavior

2.2 Manual Interview Bottlenecks

Organizations spend significant time conducting:

- Technical interviews
- Behavioral interviews
- Candidate screening
- Interview scheduling
- Candidate evaluation

Human evaluation can also introduce inconsistency and interviewer bias.

2.3 Human Proctoring Scalability

Monitoring thousands of candidates manually is:

- Expensive
- Difficult to scale
- Fatigue-prone
- Operationally complex

AssessX addresses these problems using automated monitoring with human review.

3. PRODUCT VISION

Vision

Build a trusted digital infrastructure for secure examinations and intelligent candidate evaluation.

AssessX should become more than an exam portal.

It should become an:

Assessment Integrity Platform

The system should combine:

Video

+

Audio

+

Screen

+

Identity

+

Device/session signals

+

Exam behavior

+

Answer patterns

↓

AI/Event Processing



Risk Engine



Evidence



Human Review

4. PRODUCT GOALS

Primary Goals

G1 — Secure Examinations

Provide a controlled environment for candidates to complete assessments.

G2 — Multimodal Proctoring

Detect suspicious behavior across video, audio and screen signals.

G3 — Explainable Detection

Provide evidence and reasoning rather than simply labeling candidates as cheaters.

G4 — Scalable Monitoring

Allow a small number of administrators/proctors to monitor many candidates.

G5 — Automated Evaluation

Automatically evaluate objective and coding questions.

G6 — Intelligent Interviews

Support AI-driven adaptive interviews.

G7 — Live Interviews

Allow interviewer and candidate to communicate through live video/audio.

G8 — Objective Reporting

Generate structured candidate reports and scorecards.

5. NON-GOALS

The initial product will NOT attempt to:

- Guarantee zero cheating
 - Guarantee perfect face recognition
 - Automatically punish candidates solely using AI
 - Replace all human reviewers
 - Execute untrusted code directly on the API server
 - Process every video frame using an LLM
 - Build a complete enterprise platform in the first release
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6. TARGET USERS

6.1 Student / Candidate

Uses AssessX to:

- Take examinations
 - Complete system checks
 - Verify identity
 - Complete interviews
 - View permitted results
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6.2 Admin

Uses AssessX to:

- Create examinations
- Manage candidates
- Configure security policies
- Schedule assessments
- Review suspicious activity

- Generate reports
-

6.3 Proctor

Uses AssessX to:

- Monitor candidates
 - Receive security alerts
 - Review evidence
 - Investigate sessions
 - Escalate suspicious cases
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6.4 Interviewer

Uses AssessX to:

- Schedule interviews
 - Conduct live interviews
 - View candidate information
 - Use AI interview assistance
 - Record notes
 - Evaluate candidates
-

6.5 Organization Administrator

Future enterprise role.

Responsible for:

- Organization configuration
 - Users
 - Security policies
 - Retention policies
 - Billing
 - Integrations
-

7. PRODUCT FLOW

Candidate Flow

Website



Download AssessX Windows Application



Install



Launch



Login



System Check



Identity Verification



Environment Check



Exam Instructions



Secure Exam Mode



Examination



Submission



Processing



Result / Status

8. ADMIN FLOW

Admin Login



Dashboard



Create Exam



Add Questions



Configure Exam



Configure Proctoring



Add Candidates



Schedule Exam



Monitor



Review Events



Review Evidence



Finalize Results



Generate Report

9. INTERVIEW FLOW

Candidate + Interviewer



Live WebRTC Session



Camera + Microphone



Optional Screen Sharing



AI Assistance / Proctoring



Interview Ends



Transcript / Notes



AI Evaluation



Human Evaluation



Final Scorecard

10. FUNCTIONAL REQUIREMENTS

FR-001 Authentication

The system shall support:

- Login
- Logout
- Password security
- Session management
- Role-based access

Roles:

- STUDENT
 - ADMIN
 - PROCTOR
 - INTERVIEWER
 - SUPER_ADMIN (future)
-

FR-002 Exam Creation

Admin shall be able to:

- Create exam
 - Set title
 - Set description
 - Set duration
 - Set start/end time
 - Configure attempts
 - Configure question randomization
 - Configure security policy
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FR-003 Question Management

The system shall support:

- MCQ
- Multiple select
- True/false
- Short answer
- Long answer

Future:

- Coding
 - Aptitude
 - Case studies
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FR-004 Candidate Assignment

Admin shall be able to:

- Add candidates
 - Import candidates
 - Assign exams
 - Schedule exams
 - Monitor status
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FR-005 Exam Session

Candidate shall:

- Start assigned exam
- View questions
- Navigate questions
- Save answers
- Submit answers
- Submit final examination

The system shall maintain server-side exam state.

FR-006 Exam Timer

Timer must be synchronized with server time.

The client shall NOT be the sole authority for exam duration.

FR-007 System Check

Before exam:

- Camera check
 - Microphone check
 - Screen capture check
 - Internet check
 - Permission check
 - Display check
 - Application version check
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FR-008 Identity Verification

System should support:

- Face detection
- Identity verification
- Liveness checks
- Continuous/periodic identity verification

Events include:

- FACE_VERIFIED
 - FACE_MISMATCH
 - FACE_MISSING
 - MULTIPLE_FACES
 - LIVENESS_FAILED
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FR-009 Environment Detection

System may detect:

- Mobile phone
 - Tablet
 - Additional person
 - Additional monitor
 - Book
 - Notes
 - Other prohibited objects
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FR-010 Gaze Detection

System should monitor:

- Head orientation
- Gaze direction
- Duration of deviation
- Frequency of deviations

Isolated gaze movement must not automatically generate a cheating verdict.

FR-011 Audio Monitoring

System should detect:

- Voice activity
- Multiple speakers
- Suspicious speech
- Unknown speakers

Environmental noise must be considered.

FR-012 Screen Monitoring

Where technically and legally permitted:

Monitor:

- Screen changes
- Display changes
- Application changes
- Suspicious browser activity
- Copy/paste attempts
- Unauthorized applications

FR-013 Secure Exam Mode

The desktop application should provide:

- Restricted navigation
- Exam-only environment
- Session integrity
- Application switching detection
- Display-change detection
- Relevant system restrictions

Security capabilities must be validated rather than falsely advertised.

FR-014 Proctoring Events

Every detection should generate a structured event containing:

- Event ID
 - Session ID
 - Event type
 - Timestamp
 - Confidence
 - Severity
 - Detection source
 - Evidence reference
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FR-015 Risk Score

Each examination session should have a dynamic risk score.

Suggested initial range:

0-100

Example:

0-25: Normal

26-50: Low

51-75: Medium

76-100: High

Organizations should eventually be able to configure thresholds.

FR-016 Event Correlation

The system should correlate multiple signals.

Example:

PHONE_DETECTED

+

GAZE_DEVIATION

+

UNKNOWN_VOICE

+

APPLICATION_SWITCH

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FR-017 Evidence

For important events, the system may capture:

- Screenshot
- Video segment
- Audio segment
- Screen event
- Detection metadata

Evidence retention must be configurable.

FR-018 Investigation

Admin/proctor shall be able to:

- View candidate
 - View risk score
 - View event timeline
 - Open evidence
 - Review session
 - Confirm event
 - Dismiss event
 - Escalate case
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FR-019 Reports

System shall generate:

- Exam result
 - Proctoring report
 - Integrity score
 - Suspicious events
 - Evidence references
 - Candidate performance
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FR-020 Coding Assessment

Future module supporting:

- Code editor
- Compilation
- Test cases
- Hidden test cases
- Runtime limits
- Memory limits
- Multiple languages

Candidate code must execute in isolated infrastructure.

FR-021 Live Interview

System shall support:

- Candidate video
- Interviewer video
- Audio
- Screen sharing
- Chat
- Interview timer

WebRTC shall provide real-time communication.

FR-022 Interview Proctoring

Interview sessions may use the same integrity engine to detect:

- Additional people
 - Unauthorized assistance
 - Suspicious screen activity
 - Identity changes
 - Relevant audio anomalies
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FR-023 AI Interviewer

Future feature.

AI should:

- Ask questions
 - Understand answers
 - Generate follow-ups
 - Adjust difficulty
 - Evaluate responses
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FR-024 AI Interview Copilot

Human interviewer receives:

- Answer analysis
 - Technical accuracy
 - Missing concepts
 - Suggested follow-ups
 - Interview summary
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FR-025 Candidate Scorecard

Scorecard may contain:

- Technical knowledge
- Problem solving
- Communication
- Answer relevance
- Coding
- Behavioral responses
- Integrity risk

AI-generated scores should be reviewable by humans.

11. NON-FUNCTIONAL REQUIREMENTS

NFR-001 Performance

Real-time alerts should aim for low latency.

Target:

Sub-second event notification where technically achievable.

Actual performance must be benchmarked.

NFR-002 Scalability

Architecture should support horizontal scaling.

Long-term goal:

Thousands of concurrent candidates.

No concurrency claim should be made until load-tested.

NFR-003 Availability

Production services should be designed for high availability.

Critical services should avoid single points of failure.

NFR-004 Security

All sensitive APIs require:

- Authentication
 - Authorization
 - Validation
 - Rate limiting
 - Logging
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NFR-005 Privacy

The platform must:

- Minimize data
 - Obtain appropriate consent
 - Support configurable retention
 - Secure evidence
 - Provide deletion mechanisms where required
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NFR-006 Explainability

AI alerts must provide:

- Event
 - Confidence
 - Evidence
 - Context
 - Reason
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NFR-007 Accessibility

Candidate workflows should consider:

- Keyboard accessibility
- Screen readers where applicable
- Clear instructions
- Accessibility accommodations

Security policies must not unfairly penalize accessibility-related behavior.

12. SUCCESS METRICS

Product metrics:

- Exam completion rate
- Assessment failure rate
- Proctoring detection precision
- Proctoring false-positive rate
- Detection latency
- Candidate support incidents
- Interview completion rate
- Average screening time
- Admin workload reduction

Technical metrics:

- API latency
- Event processing latency
- WebRTC connection success
- Worker throughput
- GPU utilization
- Database performance
- Concurrent sessions

13. RELEASE STRATEGY

MVP

Must include:

- Authentication
- Student/Admin
- Exam creation
- Questions
- Exam session
- Results
- Basic desktop application
- System check

V1

Add:

- Identity verification
- Camera
- Microphone
- Screen monitoring
- Basic AI proctoring
- Event engine
- Risk scoring
- Evidence
- Admin investigation

V2

Add:

- Coding assessments
- Advanced proctoring
- Collusion detection
- Live interviews
- WebRTC
- Interview scorecards

V3

Add:

- AI interviewer
 - AI interview copilot
 - Enterprise multi-tenancy
 - Large-scale infrastructure
 - Advanced analytics
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14. CORE PRODUCT PRINCIPLE

AssessX must never reduce the concept of security to:

"AI says cheating."

Instead:

Detect → Correlate → Explain → Provide Evidence → Human Review

This principle must remain throughout development.

END OF PRD